

Octahedral Units in Halide Perovskites

Yong Wang ^{1, 2 †*}, Yu Wang ^{3 †}, Tiarnan A. S. Doherty ^{4, 5}, Samuel D. Stranks ⁴, Feng Gao ³, &

Deren Yang ^{1, 2*}

1. State Key Laboratory of Silicon and Advanced Semiconductor Materials, School of Materials Science and Engineering, Zhejiang University, Hangzhou, Zhejiang, P. R. China.

2. Hangzhou Global Scientific and Technological Innovation Center, Zhejiang University, Hangzhou, Zhejiang, P. R. China

3. Department of Physics, Chemistry, and Biology (IFM), Linköping University, Linköping, Sweden

4. Department of Chemical Engineering and Biotechnology, University of Cambridge, Cambridge, UK

5. Department of Materials Science & Metallurgy, University of Cambridge, Cambridge, UK

*Email: yonwa12@zju.edu.cn; mseyang@zju.edu.cn.

†These authors contributed equally to this work.

Abstract

Metal halide perovskites with ABX_3 crystal structure possess excellent photophysical properties for (opto)electronic applications including photovoltaics, light-emitting diodes, photodetectors, and transistors. To pave the pathway toward commercial applications, enormous efforts have been made to obtain high-performance perovskite-based devices. The octahedral unit is considered to be the fundamental and functional unit of halide perovskite materials. These octahedra consist of a central B cation surrounded by six X anions, and they typically have dimensions of 5–6 Å. One promising pathway towards stable and high-performance perovskite devices is to manipulate octahedral unit in the halide perovskite samples. While rational engineering of octahedral unit has been leveraged and understood in oxide perovskites to unlock diverse functionality, analogous targeted work has been relatively underexplored in halide perovskites to date. Here, we present fundamental concepts regarding octahedral configuration in halide perovskites as well as its effects on crystal structures, photophysical properties, and device performance. We then outline techniques that can be used for characterising octahedral units and summarize different approaches to rationally manipulate octahedral unit. Finally, we discuss current challenges and the outlook for future research to align octahedral unit with device investigations. A thorough understanding of octahedral unit can facilitate modulation of structure at the atomic level, benefiting perovskite (opto)electronic devices with enhanced efficiency, stability and functionality.

[H1] Introduction

A class of semiconductors known as metal halide perovskites possess excellent physical and chemical properties including high defect tolerance and superior optical features, where production is low-cost, simple, and efficient.¹⁻⁴ State-of-the-art perovskite solar cells (PSCs) have achieved a certified efficiency of 26.7%,⁵ and their early stage commercialization process demonstrates the emergent future in the next-generation photovoltaics (PVs) area.⁶⁻⁸ The performance of perovskite-based light-emitting diodes (LEDs) has continuously improved since 2014, delivering high external quantum efficiencies over 20% for green, red, and near-infrared light, as well as decent values of 18% for blue.⁹⁻¹² These advances in perovskite PVs and LEDs have promoted the development of halide perovskites in other areas including X-ray detectors, field-effect transistors, lasers, and more.¹³⁻¹⁶ Despite the great success of halide perovskite optoelectronics, current perovskite-based optoelectronic devices still face challenges related to poor stability and bottlenecks in improving their performance. Some studies suggest that even minor manipulation of the octahedra in perovskite structures may lead to dramatically different material functionalities, indicating that manipulating the degree of octahedral tilting in the perovskite structure presents a promising avenue to further improve the performance and stability of perovskite-based optoelectronics.¹⁷⁻¹⁹

[H1] Definition and Origin of Octahedral Units in Halide Perovskites

In this Review, we focus our discussions on both conventional three-dimensional (3D) halide perovskites, and the emerging low-dimensional or mixed dimensional perovskites. 3D perovskites crystallize in the ABX_3 crystal structure, which is comprised of a cubic array of

corner-sharing BX_6 octahedra and A-site cations residing in the cuboctahedral cavities (Fig. 1a).²⁰ The formation of octahedral units in perovskites is mostly controlled by the coordination capacity of the central atoms, which can explain the differences of unit cells in many perovskite-like materials, such as the tetrahedra in Mn-based perovskites and octahedra in Pb-based perovskites. The ionic $[\text{BX}_6]$ octahedra can be considered as a super ion or atom. Just as different atoms and ions can pack to form different crystal lattices, these octahedra can, in principle, pack into different types of negatively charged, extended octahedral networks, with and without corner sharing.²¹ Halide perovskite crystal structures eventually form when these negatively charged octahedral networks are stabilized by A-site cations. In general, the variation of octahedra caused by different A-site cations can also influence a material's features.²² Such phenomena become more important when the dimensionality is lower because of the weaker interaction among metal-halide units.

The connectivity of these unit cells also matters for the material's properties. Both internal and external factors can affect the configuration of the metal-halide network. Chemical composites, such as large A-site cations, will exceed the limits of the 3D structures, causing lower dimensional perovskites with larger bandgaps, while phase transitions at different temperatures and pressures will change the connectivity of these units, leading to different charge carrier dynamics. Besides, owing to the special hybrid features in halide perovskite, the balance between hydrogen bonding and steric forces will affect the configuration of octahedral units and their array.²³ Other factors, such as electronegativity, can impact the octahedral status by altering the composite of the chemical bonding.^{24,25} As the electronegativity difference between the metal and halide increases, octahedral distortion will be enhanced including both

out-of-phase and in-phase rotations.

In the crystal structure of 3D halide perovskite, the A-site cation is typically methylammonium (MA), formamidinium (FA), cesium (Cs), or a mixture of them, B is a metal cation (typically Pb, occasionally Sn, or a mixture), and the X site is a halide anion (Cl, Br, I, or a mixture). At high temperatures, almost every ABX_3 perovskite (halide or otherwise) adopts the cubic aristotype $Pm-3m$ space group, which is a simple high-symmetry structure. The only variable within the $Pm-3m$ structure that can be modified while maintaining the space-groups high symmetry is the lattice parameter. The vast majority of perovskites are in fact distorted away from this aristotype into lower symmetrical space groups at room temperature. Of the distortions that can occur in perovskites, tilting of the $[BX_6]^{4-}$ octahedra relative to one another is by far the most common.²⁶

[H1] Modelling Octahedral Unit Variation in Halide Perovskites

The most accepted model used to define the variation of octahedral units in the perovskite family is by considering all possible octahedral tilts as combinations of component tilts around one or more of the octahedra's symmetry axes (Fig. 1b). Thus, Glazer classified the various perovskite tilt systems to understand how each one impacts crystal symmetry, ultimately identifying 23 possible octahedral tilting configurations. Later work refined this number to fifteen.^{27,28} The evaluation of distortion from the aristotype cubic perovskite phase is largely dependent on the Goldschmidt tolerance factor ($t = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)}$), where t is used to predict whether a combination of ions is able to form a cubic perovskite structure and is widely used as an indicator for stability as well.²⁹ r_A , r_B and r_X are the ionic radii of A, B and X ions,

respectively. Another important parameter, the octahedral factor (μ) is determined by the ratio of r_B to r_X in the BX_6 octahedra.³⁰ Goldschmidt hypothesized that unit cells are arranged in such a way that “the number of anions surrounding a cation tends to be as large as possible, provided that all the anions are in contact with the cation.”³¹ As a hard-sphere model, for the A cation to fit inside the cavity, the radii must satisfy the condition of $r_A + r_X \leq \sqrt{2} (r_B + r_X)$ or, equivalently, $t \leq 1$. If $\sqrt{2} (r_B + r_X) < 2r_X$, the octahedral coordination of the B cation by six X anions is not possible, and therefore $\mu \geq \sqrt{2} - 1$.³²⁻³⁴ Due to the size mismatch of the ions, halide perovskites typically possess $0.44 < \mu < 0.90$ and $0.81 < t < 1$. When t is smaller than 1 — that is, a small A cation — the corner-sharing octahedra exhibit an increasing degree of tilt, and the excessive space for A-site cations gives opportunities for deviation in the cuboctahedral cavity from the ideal rigid model (Fig. 1c). In other words, the octahedral tilting contrives to reducing the size of the A-site cavity slightly, compensating for a mismatch in the A—X and B—X bond lengths, which will be discussed later.

In order to clearly demonstrate the impacts of the tolerance factor t , octahedral factors μ and limited candidates at the A site, we calculate the different limits in the metal halide perovskites, including chemical limits with CL1, determined by the ionic radius ratio of the smallest halide anion and the largest cation, and CL2, the intrinsic reciprocal relationship between tolerance factor and octahedral factor with the smallest halide anion. Other limits such as octahedral limit (OL), stretch limit (SL), and tilt limits (TL) are also calculated and shown in **Fig. 1d**, the blue area of which is the only regime providing geometrically stable cubic-like structure.³¹

Similar to other members in the perovskite family like perovskite oxides, the tilt is

normally dominated by the steric effect of the A site cation occupying the interstitial, central cation site of BX_6 octahedra, in which the mismatched sizes of the A and B ions induce octahedral tilting due to steric repulsion.³⁵ In $MAPbI_3$, to reduce the tilt of the octahedra, it is useful to introduce appropriately sized A-site cations to enable better stability in the tetragonal phase.³⁶ Beside the steric effects, the hybrid composite in halide perovskites also enable hydrogen bonding interactions between the A-site organic cation and inorganic frame, playing a vital role in manipulating the configuration of the octahedra.³⁷⁻³⁹

The ideal rigid-ball model, and the analysis based on single octahedra, is straightforward and provides an intuitive prediction of the crystal structure and chemical composite. With greater efforts, such as by considering the overall effects (including hydrogen bonding, steric effects, electronegativity), the correlation between perovskite properties and lattice can be understood. Wiktor et al assessed the local octahedral tilting and phase transitions in a perovskite structure and demonstrated the potential inhomogeneity of phases in the studied samples, even in the case of the cubic phase above the cubic-to-tetragonal transition temperature.⁴⁰ Nevertheless, after analysing the orientation of the A-site molecules, Zhang et al demonstrated the dependence of radiative recombination on the A-site molecules' direction, indicating that an accurate picture of the lattice structure is necessary to understand the correlation between a structure and its physical features.⁴¹

Such variations of octahedral units in halide perovskites impact the photophysical properties and structural stability of halide perovskites as shown in reports on band structures,^{17,42} carrier dynamics,^{43,44} and even the phase and chemical stability of materials and devices^{45,46}. Therefore, manipulating octahedral units present a new avenue to further stabilize

the structure and apply the excellent features of perovskite materials in the real applications. Such a rearrangement of soft crystal structures is possible to adapt carrier dynamics and enhance the performance of perovskite-based devices as well.^{19,23,47-50}

[H1] Characterization of octahedral unit in halide perovskites

Characterizing octahedral tilting in halide perovskites can be challenging, particularly when the tilt is minor, occurs only locally (such as spatially heterogenous), or is dynamic in nature. On the other hand, the changes of octahedral units can manifest in various forms, such as macroscopical stress and strain.

[H2] Diffraction techniques

The most intuitive way is to use diffraction techniques as they are suitable for identifying crystal structures and can demonstrate the octahedral configuration. As a lowering of symmetry manifests in the appearance of additional sharp Bragg peaks or diffuse scattering in diffraction patterns that can be assigned to the superstructure formed when tilting away from the aristotype cubic $Pm-3m$ structure (Fig. 2a).⁵¹ Single crystal X-ray diffraction (XRD) can be used to obtain quantitative information on the octahedral unit in a halide perovskite material by inspecting the intensity of these superstructure peaks and refining the diffraction data to a structural model.⁵² However, due to the relatively weak interaction of X-rays with matter, superstructure peaks are not always of sufficient intensity to be resolved above the level of noise if the unit cell distortion is small. Additionally, traditional XRD experiments on perovskite thin-films provide an ‘average’ rather than a ‘local’ structural picture as many grains in a polycrystalline film are

sampled, which may mask local octahedral tilting distortions.⁵³ This can be mitigated through experimental design, for example by the use of local synchrotron X-ray probes that can allow individual grains to be resolved or by performing single crystal measurements with synchrotron radiation.^{51,54} Different from traditional XRD, X-ray diffuse scattering can be used to estimate the dynamic local octahedral tilting by focusing more on the disorder in the materials.⁵⁵ Neutron diffraction is another powerful technique that can provide quantitative information on octahedral tilting in perovskite materials and is more sensitive to subtle structural distortions than XRD. However, neutron diffraction requires the use of very thick single crystals or powder samples, which may be impractical and whose properties may deviate from thin-film halide perovskite analogues. Inelastic scattering experiments with either X-rays or neutrons can help resolve whether observed symmetry breaking peaks are due to a local or macroscale distortion, if the distortion is dynamic or static in nature, and provide detailed insights into lattice dynamics.^{51,56-58}

Electron diffraction (ED) is exceptionally sensitive to subtle symmetry breaking due to the strong Coulombic interaction of electrons with matter. It has been widely used to interrogate octahedral tilting in oxide perovskites and in halide perovskites, where it revealed that A-site cation-alloying influences the magnitude of octahedral tilting.^{19,59} As the electron beam can be focused to sub-nanometer sizes and the electron scattering cross-section is very large, high signal-to-noise single-crystal diffraction patterns can be extracted from small individual grains in halide perovskite thin films at low electron doses (Fig. 2b). This makes ED a particularly powerful technique for characterising octahedral tilting, even when the distortions manifest over short length scales or are small in magnitude. Additionally, ultrafast electron diffraction

experiments can provide insights into transient octahedral tilting or other structural distortions.^{60,61} Care must be taken, however, when analysing ED patterns: the same strong interaction between electrons and matter that makes ED a powerful technique, also gives rise to complex multiple scattering events and dynamic interactions between the scattered and incident electron waves, particularly when an electron beam interacts a sample of thicknesses exceeding ~ 100 Å.⁶² These ‘dynamic interactions’ can manifest as the appearance of forbidden reflections in ED patterns and cause observed diffracted intensities to deviate substantially from those predicted by the kinematical diffraction theory, which is commonly used when analysing XRD patterns. An example of this can be seen in Fig 2b where the experimentally observed intensity of the $01\bar{1}$ reflection in an ED pattern extracted from a FAPbI_3 perovskite with 10 mol% additive of CsCl deviates from the simulated intensity according to kinematical theory — this difference is in part due to dynamic effects. Beam-induced damage can also complicate the analysis of ED patterns extracted from halide perovskite materials.^{63,64}

[H2] Atomic resolution imaging

Beyond diffraction, other techniques can also provide direct atomic resolution imaging, such as transmission electron microscopy (TEM) or scanning transmission electron microscopy (STEM). In other perovskite families, especially perovskite oxides, (S)TEM has been used to characterise octahedral tilting in different ways (mostly bright field and dark field, Fig. 2c-2f). However, it is challenging to resolve octahedral tilting of small magnitudes and fully decouple these signatures from image artefacts, and real space techniques such as these can only resolve tilts in the plane perpendicular to the incident beam direction.⁶⁵ Additionally, atomic resolution

STEM imaging requires relatively high electron doses and care must be taken when imaging beam sensitive materials, such as halide perovskites, but it is possible to obtain atomic resolution images of halide perovskites without excessively damaging the material.⁶⁶

[H2] Other analytical techniques

Nuclear magnetic resonance (NMR) has also been successfully used to resolve octahedral tilting and phase transitions in halide perovskites, though in some cases it has not resolved subtle signs of symmetry breaking that can be resolved with nuclear quadrupole resonance (NQR) and/or ED.⁶⁶ Compared to above XRD, Raman spectroscopy is a more local probe, which can also be a powerful technique for probing lattice dynamics and the local environments at the microscopic scale, though it typically only probes symmetry allowed optical modes at the Brillouin zone centre.⁶⁷ Additionally, optical spectroscopy can also indirectly indicate the tilt of the octahedra through changes in its optical properties.

[H1] Octahedral units and lattice dynamics

Though the hard-sphere model outlines the geometric limits of octahedral configurations, halide perovskites are also well-known for their soft nature, indicating that octahedral tilts and deformation occur among lattices. The octahedral tilt in perovskites is usually considered as a deviation from the rigid hard-sphere model without the deformation of the unit shape, caused by interactions among composites (such as steric effects and hydrogen bonding), or caused by the strain in the material.⁴⁸ In addition to the focus of local octahedral unit, the comparison with the information of the averaged information in a larger scale is important as well. Wiktor

et al found that even in the high-symmetry structure (tetragonal or cubic) from the global view, even in the high-symmetry structure (tetragonal or cubic) from the global view, octahedral orientations can still be tilted with a standard deviation of up to 7° in each direction locally at high temperatures in the system of CsPbI_3 .⁴⁰ Moreover, experimental evidence has corroborated this theoretical result, demonstrating that in various perovskite compositions, localized area can exhibit lower symmetry, while the bulk phase maintains higher symmetry.⁵⁵ Such differences are crucial for analysing the distinctions between theoretically and experimental band structures. Different from the assumption of the A-site composite central position, the element or molecule can also deviate from its geometric centre. Strikingly, Bechtel et al showed that it is vital to maintain the *Pnma* phase with the help of the interactions between octahedral tilting and A-site displacements.⁴⁸ As for low-dimensional perovskites like 2D perovskites, tilting can be greater due to the balance between steric and hydrogen bonding interactions, leading to the considerable inversion asymmetry and impact on the charge carrier dynamics.²³

Instead of its standard shape, octahedra can also deform into different shapes and it is also important to evaluate the influence of such variations on the lattice dynamics. Zhu and co-workers suggested that a distorted crystal structure, caused by octahedral deformation, can impact stereochemical expression (SE) of the ns^2 electron pair (NSEP) on group IV metal cations — which can be considered as one of the main reasons for lattice instability and broken inversion symmetry.⁶⁸ The inherent lattice instability associated with NSEP indicates that these crystal structures (even the stable centrosymmetric structures) are susceptible to atomic displacements under external stimuli, such as heat and electric fields (Fig. 3a). The lone pairs

of electrons in halide perovskites lead to a considerable destabilization of centrosymmetric structures that are favoured by electrostatic forces. Therefore, adopting static or dynamic distorted structures depends on the minimal energy of these two cases (Fig. 3b). This competition leads to various potential energy surfaces (PES), which are determined by the size of the A-site cation, the strength of cation-anion interaction, and the structure's dimensionality. For the undersized A-site cations, the PES of B-site cation displacement is fairly flat at the bottom of the potential (Fig. 3b, green). As the A-site cation size increases, elongated B–X bonds tend to optimize the covalency, leading to several stable but off-center positions for the B cation, and finally breaking the cubic structures and then reconfiguring the chemical bonds. If the potential barrier between two minima is high enough (for example, larger than the thermal energy), a double-well potential can be kept in the crystal which leads to static distortion (Fig. 3b, yellow). Otherwise, a low potential barrier is unable to anchor B-cations and such delocalization leaves the ions dynamically distributed at multiple minima (Fig. 3b, purple). In short, PES with flat bottoms or potential barriers lower than the thermal energy are expected to result in large atomic displacements. Nevertheless, larger distortion under thermal expansion can be attributed to bond length elongation.

The dynamics of lone-paired electrons result in anharmonic lattice vibrations and dynamic disorder, observed in the temperature-dependent, low-frequency Raman spectra of CsPbBr₃ (Fig. 3c), MAPbBr₃ (Fig. 3d), and FAPbBr₃ (Fig. 3e).⁶⁹ At high temperatures, cubic phases without any octahedral variation exhibit similar broad central peaks near zero frequency, suggesting liquid-like polar fluctuations. When the temperature is decreased, tetragonal phases with small octahedral tilt exhibit less diffusive spectra, and the central peaks remain. As

octahedral tilts become larger in the orthorhombic phase, their spectra become fundamentally different. The orthorhombic phases of CsPbBr₃ and MAPbBr₃ show well-resolved vibrational modes from the inorganic sublattice, while orthorhombic FAPbBr₃ maintains a central peak despite narrower linewidth. This is different from the previous understanding that the local polar fluctuations in CsPbBr₃ are assigned to a head-to-head Cs⁺ motion coupled to Br⁻ face expansion.^{69,70} Asbury and co-workers characterized the changes of anharmonic lattice vibrations in perovskites and revealed the influence of these changes on carrier recombination and ion transport by comparing vibrational behaviors when the carriers are excited or not.⁷¹ These results indicate that manipulating phonon dynamics in perovskite lattices by controlling octahedral tilting may lead to novel strategies for fine-tuning perovskite properties, especially charge carrier dynamics and ion migration.

[H1] Octahedral unit and bandgaps of halide perovskites

The bandgap of halide perovskites is one of the most important features for their optoelectronic applications including solar cells and LEDs. Interestingly, many halide perovskites show the feature of direct bandgap structures, implying that their devices can both benefit from large absorption coefficients for solar cells and remarkable radiative attributes for luminescent devices. Moreover, the sharp band edge also gives them decent carrier transport properties because of the relatively low effective masses and low effective density of states for the lasing actions.^{72,73}

The absorption features are used to evaluate the potential in PVs since theoretical predictions reveal that the best potential efficiency of single-junction solar cells is 33.7% with

a 1.35 eV bandgap semiconductor,⁷⁴ while radiative properties are more prominent in the LED arena for better device design and wider colour gamut. Undoubtedly, a suitable bandgap is vital to achieve highly efficient perovskite-based optoelectronic devices.

Despite various approaches toward tuning bandgaps of mixed-halide perovskites, the simultaneous requirements on both performance and stability require alternative approaches related to lattice dynamics including A-site cation synthesis, composition combination, and strain engineering, all of which have been developed to balance the bandgap, performance, and stability of halide perovskites. Chen and co-workers found that strain in halide perovskites influences the hybridization of bonding and anti-bonding orbitals, leading to adjustments in bandgaps (Fig. 4a).⁷⁵ Based on the case analysis of compressive, to zero, to tensile biaxial strain, the expansion of the in-plane lattice weakens the Pb-X bonds and thus in principle pulls down both the anti-bonding valence band (VB) and conduction band (CB) energy levels. However, the VB with strong anti-bonding hybridization is more substantially decreased. As a result, the bandgap increases from compressive, to zero, to tensile strain.⁷⁶ The control of a crystal lattice in certain directions suggested that substantial improvements were achieved by controlling the octahedral configuration as well as consequent hole carrier transport and extraction across the interface of perovskite absorber/hole transport materials.

From another perspective, managing octahedral configurations can be considered as a more general pathway towards regulating bandgaps, as the variation of octahedral states indicate the changes on the Pb-X-Pb angle, off-centering of Pb atoms in the PbX₆ octahedra, and Pb-X bond length — all of which affect the electronic structure of the perovskite.⁷⁷ The VB maximum of APbX₃-type lead halide perovskites corresponds to the antibonding hybrid

state of the metal s and halide p orbitals, while the CB minimum is the hybrid of metal p and halide p orbitals with less antibonding character (Fig. 4b).¹⁷ Minor changes in the perovskite lattice increase the amount of metal-halide overlap, altering the band structure. That is to say, the conduction bands derived from the p orbitals, having poor orbital overlap, are expected to respond less strongly to distortions in the lattice structure than the valence band that is derived from metal s orbitals. Therefore, the variation of octahedral units that boosts metal-halide overlap will decrease the bandgap, whereas the variation of octahedral tilting that reduce metal-halide overlap will increase the bandgap. In all, the deformation of the octahedra including compression (or contraction) and anharmonic variation changes the potential distribution contributed by atoms, leading to a shift of the bandgap.

As found in MAPbI₃ systems, partial replacement of MA⁺ with FA⁺ causes a phase transition from tetragonal to a quasi-cubic structure due to the interplay of the A-site cation size, enhanced hydrogen bonding between the FA⁺ cations, and the inorganic matrix, altering the covalent or ionic character of Pb–I bonds.⁴² The variation of octahedral tilt caused by phase transitions provides an opportunity to tune the bandgap, which is also supported by DFT calculations. The decrease of the bandgap by the substitution of the organic cation (MA→FA) in MAPbI₃ is attributed to the suppressed octahedral changes. Additionally, octahedral tilts also lead to major changes in the carrier effective masses, potentially enhancing electron and hole transport in FAPbI₃ perovskite compared with the tetragonally distorted MAPbI₃.

Different from organic cations, the small isotropic Cs⁺ cation is free from hydrogen bond or orientation effects, making Cs-based inorganic perovskites (especially CsPbI₃) special with respect to various properties including crystal structure distortion and other photophysical

features. The state of octahedral units in CsPbI₃ depend on strain remnants in the structure. Therefore, the bandgap of CsPbI₃ reported by different groups shows subtle differences. Fig. 4c and 4d show the effect on the bandgap of octahedral rotation and Pb off-centering in CsPbI₃.⁷⁸ Both VB and CB are anti-bonding states (Pb-6s – I-5p at VB and Pb-6p – I-5p at CB). As the angle of octahedral rotation increases, the elongated Pb–I bond weakens the Pb–I coupling, which results in a downshift of VB and CB energy levels. However, the VB downshifts more than the CB as the VB is more sensitive to the lengthened bond due to inner-shell Pb-6s states contributing to the VB. The relatively large downshift of VB leads to the bandgap blueshift in CsPbI₃. Moreover, such understanding can also be extended to the mixed-dimensional perovskites. By introducing extra ethylammonium cations, the emission peak of the 1-naphthylmethylanmonium iodide-passivated CsPbI₃ blueshifts in contrast to the redshift caused by the lattice expansion.⁷⁹

[H1] Octahedral unit and carrier dynamics in halide perovskites

Octahedral unit, as fundamental component of halide perovskites, plays a crucial role in governing carrier dynamics, including carrier generation and recombination processes, as well as phonon modes and configurations.

[H2] Effect of octahedral variation on carrier generation

High absorption coefficients in different perovskites are widely investigated and applied in solar cells and photodetectors. The Shockley–Queisser (SQ) limit, approximately 33%, defines

the maximum energy a solar cell can extract from a photon. This limit underscores the outstanding efficiency potential (exceeding 66% power conversion efficiency, PCE) by using hot carriers, in contrast to conventional devices.⁸⁰ In other words, the generated hot carriers do not work efficiently in photovoltaic devices before they cool down to the band edge, so the carrier generation in the devices at thermal equilibrium also covers carrier cooling processes. Perovskites manifest a very slow carrier cooling process, the rate of which is several orders of magnitude lower than conventional absorbers such as silicon. This slow cooling theoretically enables highly efficient hot-carrier extraction (up to $\approx 83\%$).⁸⁰ As thermalization processes need the assistance of phonons, a phonon bottleneck is usually considered as the main reason for the possibility of making use of hot carriers, when limited well-matched phonons will be barely involved in the cooling processes, as shown in Fig. 5a. With pressure-dependent transient absorption spectroscopy, it was found that after the insulator-to-metal Mott transition, the cooling process can be accelerated with higher pressure due to the enhancement of electron-photon couplings (Fig. 5b).⁸¹ According to calculations, this variation can raise the dissipation energy from carriers to the lattice through specific phonon modes ($\sim 27\text{ cm}^{-1}$, $\sim 0.8\text{ THz}$) in perovskites, which is the only mode showing pressure sensitivity.⁸¹ More importantly, this mode is related to both the octahedral twist and distortion, and the electrons at the bottom of the CB, implying that slower hot-carrier cooling processes might occur due to the weakening of local electron-phonon coupling.

Other structural traits in the halide perovskites are also crucial and affect the performance of materials and devices through their octahedral states. Ionic-like connections between constituent anions and cations — different from the covalent bond in many materials like

silicon — means anharmonic reshapes of the octahedra can also change spacing between the centres of cations and anions. This is also seen for the spontaneous polarization in the III-V group materials.⁸² When it comes to the thin-film layer, dynamic configurations like dangling bonds are abundant at the surface of perovskite materials, paving a facile pathway to forming surface dipoles and varying the band alignment in the heterointerfaces. More importantly, this sheds light on an uncommon doping method; polarization doping.^{83,84} Induced carriers (generated through the polarization field) and surplus space charges (net charges caused by mismatched cations and anions), can also affect the properties of materials and device performance. These charges can be manipulated by octahedral tilting as the structural distortion (such as surface configuration) may screen or strengthen the electrostatic effect from surplus space charges.

[H2] Effect of octahedral variation on carrier recombination

Carrier recombination mechanisms play a fundamental role in different types of semiconductor devices. Taking the example of solar cells, carrier recombination limits charge extraction in active media, which is why better solar cells have materials with slower carrier recombination rates. For simple transient processes, carrier density $n(t)$ is time-dependent and can be described by different recombination processes, as expressed through the rate equation:

$$\frac{dn}{dt} = G - R = G - k_1n - k_2n^2 - k_3n^3 \quad (1)$$

here, R is the overall recombination process, k_1 , k_2 and k_3 are three rate constants mainly associated with three different recombination processes: trap-assisted recombination; radiative recombination; and Auger recombination, and G represents the generation rate of charges from,

for example, photoexcitation. Through trap-assisted recombination, carriers relax into ground states, with the carrier energy lost to heat and lowers the conversion efficiency regardless of the type of solar cells or LEDs. Radiative recombination is an efficient way of generating photons. Auger recombination is a many-body process where the energy generated by electron-hole pairs is transferred and excites another carrier into higher excitation in a non-radiative way⁴⁴ — such processes are mostly relevant at high carrier densities and therefore has a considerable impact primarily on LEDs or lasers. At steady state, carriers mainly present a temporal and spatial distribution. Thus, to present the overall influence of carrier recombination and carrier transport, we can evaluate the result with the help of the Einstein relationship:

$$r(n) = k_3 n^3 + k_2 n^2 + k_1 n, \quad L_D(n) = \sqrt{\frac{\mu k_B T}{r(n) e}}, \quad (2)$$

where μ is the carrier mobility, T is the temperature, k_B is the Boltzmann constant, and e is the elementary charge, L_D is the diffusion length, $r(n)$ is the recombination rate.^{85,86}

Device physics places straightforward demands on the materials. As discussed in the previous part, the change of octahedral configuration in the perovskite has notable effects on the band structure, not to mention the substantial impact on carrier dynamics, especially recombination and generation as well as the corresponding carrier diffusion length. In addition to the well-described Shockley-Read-Hall (SRH) relationship on trap-assisted recombination, radiative recombination tends to decrease at higher temperatures while Auger recombination relies more on the specific phases. Therefore, the carrier diffusion length gradually decreases from about 3 μm to 1.2 μm along with a series of rising temperatures (from -93°C to about 67°C), corresponding to two specific phases (tetragonal and cubic).⁸⁷

In addition to thermal effects, strain can also vary the configuration of octahedra and

excited states which is also found in other semiconductors. In the FAPbI₃ perovskite, tensile strain elongates the Pb-I bonds and enhances structural fluctuations (suppressing the electron-hole radiative recombination).⁸⁸ The lattice-level distortion elongate carrier recombination lifetime to 1.1 and 3.1 of its original value through lattice contraction and octahedral tilt.⁸⁹ By comparing the various processes involved in carrier dynamics, it can be identified the important influence of lattice contraction and octahedral tilting on both radiative and non-radiative recombination, mediated through non-adiabatic coupling rather than dephasing. This suggests an indirect correlation between A-site composition and recombination processes. Later work also pointed out that such coupling relates the trap-assisted recombination with different defect types and lattice vibration.⁹⁰

In fact, the contribution of symmetry breaking has been widely investigated in many aspects, such as the Rashba effect (spin splitting in the band structure) and abnormal Auger recombination (resonance in the band structure). Ghosh et al. suggested that the distortion of surface terminations break the centrosymmetric crystal structure and enhance the Rashba effect — which splits the band structure in both momentum and energy space, and leads to a direct-to-indirect band gap transition (Fig. 5c).⁹¹ Their further results show that such variations may reduce carrier recombination in these mixed-cation perovskites. Furthermore, the intra-band structure caused by the Rashba effect can induce strong Auger recombination due to the energetic resonance, relieving the strict limitation of momentum conservation in the Auger process (Fig. 5d).⁹² Studies have proposed that the structural distortion also enables the formation of bright excitons and, more importantly, that band splitting can be achieved by the temperature-induced lattice distortion of perovskite nanocrystals. This demonstrates the

unpredictable potential of perovskites in quantum communications and computing with the support of controllable crystal structure after the fabrication processes.⁹³ Different from 3D perovskites, the variation in lower dimensional perovskites complicate the influence on the recombination process, as it is also related to the nearby octahedral units. The management of a four-Pb-shaped quadrilateral can change the emission linewidth by controlling the sizes of excitons and self-trapped excitons.²² Furthermore, surface octahedral tilts in layered perovskites will induce an excitonic redshift in the luminescent process, and promotes Rashba spin splitting in few-layered perovskites.²³ All of these results indicate the importance of manipulating octahedral units for desirable charge carrier dynamics in different perovskite optoelectronic applications.

[H2] Effect of octahedral variation on electron-phonon coupling

Octahedral tilting is not only important in carrier dynamics (recombination and generation), but also changes phonon modes and configurations (such as coupling and oscillation strengthens) as well as emission linewidth, thermal conduction, and carrier mobilities. Herz and coworkers have conducted a throughout analysis of carrier–phonon interactions in different perovskites (FAPbI₃, FAPbBr₃, MAPbI₃, and MAPbBr₃).⁹⁴ They found that the Fröhlich interaction between charge carriers and (Longitudinal optical) LO phonons provides the dominant contribution to the predominantly homogeneous linewidth broadening in these hybrid perovskites at room temperature. Moreover, Fröhlich interactions are stronger in bromide-based perovskites than iodide-based perovskites, connecting material composition to the carrier-phonon coupling as one of the most fundamental scattering processes during charge

carrier transport. Normally, electron-phonon coupling is presented in two major channels in polar semiconductors through deformation potentials (lattice distortion), and electromechanical or piezoelectric interactions (induced electric fields). The former changes the electronic band structure while the latter modifies the electronic Hamiltonian, both of which can control carrier transport via scattering (Fig. 5e). For example, Sargent and co-workers added longer organic ligands into a 3D perovskite crystal and managed to regulate the octahedral unit, and as a result, the reduced electron-phonon interactions boost non-radiative decay and result in lower photoluminescence quantum yield.⁹⁵ Also, the strength of electron-phonon interactions is related to the octahedral variation level. In different crystal phases of MAPbI₃ perovskites, electron mobilities vary in the range of 5–10 cm² V⁻¹ s⁻¹ and hole mobility within 1–5 cm² V⁻¹ s⁻¹.⁹⁶ Many-body ab initio calculations demonstrated that LO phonons at low frequencies should be the main reason for low mobilities in perovskites. Interestingly, for hybrid or inorganic perovskites, the corresponding states of octahedral units are quite different, emphasizing the importance of octahedral regulation (Fig. 5f).⁹⁷ Other works also confirmed that these different mobilities of carriers in halide perovskites are strongly related to the electron-phonon coupling strength caused by octahedral tilting.⁹⁸⁻¹⁰¹ In addition to carrier-phonon coupling, polarons have also been proposed as an alternative method of impacting carrier mobilities.¹⁰² Therefore, establishing how charge carriers interact with the lattice in these halide perovskites is essential for developing high-efficiency perovskite optoelectronics. Compared with bulky materials, excitonic behaviour in confined systems will not only become more important but also be affected more seriously by the phonon contributions, that is lattice dynamics of structural variations. In the exfoliated quasi-2D perovskites, PL redshift at higher

temperatures can be highly suppressed by the hexagonal boron nitride cover, which is attributed to the enhanced ϵ .

[H1] Octahedral unit and stability of halide perovskites

Halide perovskites often exhibit phase transitions under certain conditions. Undesirable phase transitions have a considerable effect on a material's electronic structure and optical properties, which are considered to be one of the main reasons for the stability challenge for both FAPbI₃ and CsPbI₃ perovskites.^{103,104} For example, CsPbI₃ phase transitions involve the octahedral rotation along the [001] axis. When a corner-sharing octahedron rotates along the [001] axis, the corner-sharing octahedral structure is broken, forming a non-absorbing face-sharing octahedra structure.^{105,106} Therefore, manipulating and stabilizing octahedral units is critical for achieving highly stable perovskites.

Initially, alloying Cs⁺, MA⁺, or both, have been adopted to grow the cubic phase of FAPbI₃-rich perovskites, thus stabilizing the photoactive FA-based halide perovskites for efficient photovoltaics.¹⁰⁷⁻¹⁰⁹ Especially during the decomposition process of organic A cations in hybrid perovskites, the loss of ions leads to a failure in the maintenance of the pristine perovskite Pb-I framework. However, the degradation pathway is dependent on the conditions of the A-site cations (type and occupancy).¹¹⁰ Stranks and co-workers revealed that stabilized FA-rich perovskites are actually not cubic and exhibit ~2° octahedral tilt at room temperature.¹⁹ The tilted octahedra in these tetragonal *P4/mbm* FA-rich perovskites provide an innate barrier against a hexagonal face-sharing structure (Fig. 6a,b). According to this principle, ethylenediaminetetraacetic acid with a large steric bulk is introduced to template the growth of

octahedral-tilt-stabilised (ots)-FAPbI₃ thin films without cation alloying. As such, the photoactive ots-FAPbI₃ is extremely stable against external stresses such as temperature, humidity, and light when compared to cubic α -FAPbI₃. Theoretical results suggested that the stabilization energies, that is the energy differences between cubic and the most stable tilted phases, are linearly correlated with the tolerance factor, t (Fig. 6c), and the moderately tilted phases are thermodynamically more stable than their regularly cubic parents.¹¹¹

In addition to the case of FAPbI₃, the photo-active black phase of CsPbI₃ with a non-ideal tolerance factor was considered to be unstable at room temperature.^{103,112,113} Studies have demonstrated that the thermodynamic instability of the γ -phase of CsPbI₃ arises from three factors: Cs rattling in its coordination polyhedron; low coordination number of Cs-I; and a high degree of Pb-I octahedra tilting.¹¹⁴ As the degree of [PbI₆]⁴⁻ octahedral tilt can profoundly affect the transition between different phases, black-phase CsPbI₃ is favoured when the octahedra tilt is controlled by introducing steric hindrance or other external factors (Fig. 6d).^{50,115,116} Wang et al managed to reduce octahedral tilting and successfully created the tetragonal (β) phase of CsPbI₃ with suppressed tilting in comparison with γ -CsPbI₃ (Fig. 6e). The β -CsPbI₃ phase exhibits excellent stability against light and temperature,¹¹³ and PSCs with this phase have displayed PCEs exceeding 20%.^{117,118} These achievements demonstrate that perovskite stability can be extended by suitably tilting octahedral units in the crystal both internally and externally. Another popular approach is to employ mixed cations such as Cs and formamidinium (FA) combinations. But theoretical calculation conducted by Yin et al pointed out that the most efficacious status is to uniformly distribute Cs⁺ cation in the framework so as to maximize the octahedral tilting and the system stabilization.¹¹⁹

[H1] Manipulating octahedral units in halide perovskites

Manipulating octahedral tilting to achieve desirable properties has been successfully carried out in the oxide perovskite community.^{120,121} Several promising approaches are currently being explored in halide perovskites including compositional engineering, growth kinetics-associated modification, and interfacial treatments.¹²²

[H2] Composition engineering

Compositional engineering involving substituting different ions into the structure is a method of manipulating octahedral units in halide perovskites (Fig. 7a, b). It is well accepted that the ionic radius can determine the tilting degree in halide perovskites,^{38,106} and according to the previous discussion on the octahedral unit, these ions — once inside the structure — have dual functions: geometrical effects and chemical interactions. Typically, the introduction of mixed ions at A, B, or X sites will change the steric hindrance and therefore tilt the octahedra. For example, the increase of halide sizes leads to a greater degree of octahedral tilt and expands the lattice.¹⁰⁶ This is because the enlarged cavity within the inorganic octahedral metal-halide framework is unsuitable for the A-site cations. As a result, the octahedra tilts to reduce the cavity volume as halide size increases.¹²³ When it comes to A-site molecules, results are harder to predict.^{87,107,124,125} In general, reducing the A-site cation size increases the degree of octahedral tilting. The organic species can squeeze into the inorganic framework, and dictate the tilt of BX₆ octahedra, and cause structural distortions leading to deviations from the ideal X–B–X bond angles.¹²⁶ For example, the incorporation of ethylammonium (EA) in MA-based

perovskites greatly stretches Pb–I bonds, expands the cage, and induces a larger octahedral distortion in the inorganic framework (Fig. 7c).¹²⁷

In addition to the steric effects that can impact the degree of tilt in halide perovskites, chemical interactions — especially hydrogen bonding between organic A-cations and halides — is also of great importance. In the FA-Cs mixed system, hydrogen bonding can be varied through the different ratios of Cs and FA, differing the interaction between the organic species and halides in the PbX_6 octahedra. This results in $\text{Cs}_{0.17}\text{FA}_{0.83}\text{Pb}(\text{Br}_{0.4}\text{I}_{0.6})_3$ with a cubic perovskite structure and $\text{Cs}_{0.4}\text{FA}_{0.6}\text{Pb}(\text{Br}_{0.17}\text{I}_{0.83})_3$ with a tetragonal structure of $P4/mbm$ space group.¹²⁸ In contrast to MAPbI_3 with an out-of-phase octahedral tilting pattern, $\text{Cs}_y\text{FA}_{1-y}\text{Pb}(\text{Br}_x\text{I}_{1-x})_3$ perovskites have an in-phase pattern of octahedral tilting about the c-axis. This symmetry gives rise to specific reflections in the XRD patterns for tilting angles and lattice constant calculations at different temperatures across the transitions. Furthermore, efforts can be taken by modulating these two approaches spontaneously. In general, substitution of low concentrations smaller cations, such as, Cs^+ and Rb^+ , at the A site in FAPbI_3 results in a global ‘locking’ of the $[\text{PbI}_6]^{4-}$ octahedral tilting.⁴⁷ The oscillation of the octahedral tilting in these FAPbI_3 materials is therefore greatly reduced, indicating that the lattice dynamics are reduced when the octahedral framework is ‘locked’. Using a smaller cation, such as Cs^+ , can tilt the BX_6 octahedra in the tetragonal structure, reducing the X–B–X bond angle and increasing band gap.¹⁷

In short, compositional tailoring has been successful in manipulating the octahedral tilting and achieved many remarkable examples to enhance device performance. However, delicate adjustments of multiple cation compositions are crucial, otherwise improper combinations can

be detrimental to the long-term stability of films and devices.^{108,129} Additionally, homogeneously mixing A-site cations across an entire perovskite film during solution processing can be exceptionally challenging and may facilitate the formation of local regions of the film that further compromise long-term performance and stability — for example, through the formation of hexagonal phase impurities in FA-rich films.^{19,130}

[H2] Growth kinetics-associated modification.

Crystal growth is always an important topic for any kind of semiconductor material. For halide perovskites, growth kinetics can be more vital for the manipulation of octahedral units (Fig. 7d). Pressure methods are used to introduce extra pathways such as amorphization and recrystallization for the high-quality MASnI_3 with larger octahedral tilting compared to lead-based perovskites, resulting in enhanced structural stability.¹³¹ In general, the degree of octahedral tilting increases under higher pressure because the octahedra are more compressible than the lattice.¹³² Additionally, the surface-guided chemical vapor deposition method has also been adopted to successfully fabricate CsPbBr_3 nanowires.¹³³ Both nanobeam scanning electron diffraction and geometric phase analysis indicate that these nanowires exhibit large but uniform lattice rotations. Such phenomena should stem from heteroepitaxial mismatch, accentuated by a considerable difference between the thermal expansion coefficients of the sapphire substrate and the CsPbBr_3 crystal, offering the chance to impact octahedral tilt.

In general, ligands and additives can influence both nucleation and growth processes in different aspects, including the chemistry in the precursors, the energies at interfaces, and also an enhanced stabilization effect during crystal growth.^{113,134,135} Ligands can act as capping

agents, controlling the size and shape of perovskite nanocrystals by selectively binding to certain crystal facets.¹¹⁵ This selective binding can lead to the preferential growth along specific crystallographic directions, influencing the arrangement of octahedral units. Additives such as surfactants or complexing agents can also alter the nucleation and growth kinetics, leading to changes in crystal morphology and arrangement. For example, organic cations can be introduced into the CsPbI₃ precursors and make use of the intermediate phase similar to DMAPbI₃ perovskite, resulting in the formation of the β -CsPbI₃ perovskite with small octahedral tilting in comparison with γ -CsPbI₃ perovskite.¹³⁵ These inspiring results highlight the possibilities of regulating octahedral tilting at different stages such as during templating crystallization at the beginning stage, vapor-controlled processes in the growth stage, and pressure-induced recrystallization during post-treatment.

[H2] Interfacial treatments

Interfaces of halide perovskites have a great impact on the structural features due to the ionic nature compared with conventional semiconductors. In other words, tailoring interfaces can effectively manage octahedral tilting (Fig. 7e). In previous work about the strained epitaxy of halide perovskite single-crystal thin films on lattice-mismatched halide perovskite substrates, residual strain effectively changes the octahedral tilts, leading to a reduced bandgap and increased hole mobility of FAPbI₃ perovskites.¹³⁶ Steele et al. used substrate clamping and biaxial strain to induce the PbI₆ octahedral tilt and achieve stable black-phase CsPbI₃.¹³⁷ Such interfacial management can stabilize the perovskites via fixed octahedral tilting. Furthermore, introducing interfacial layers with organic groups can also regulate octahedral tilting in

different pathways. For example, incorporating EDAPbI₄ perovskites into the γ -CsPbI₃ grain boundaries can regulate the octahedra tilts and result in the black phase CsPbI₃ with enhanced stability.¹³⁸ Subsequently, the aromatic quaternary ammonium salt is deposited onto the surface of γ -CsPbI₃. As such, the aromatic quaternary ammonium PTA⁺ cations attach to the interfacial lattice, with Br⁻ inserted in the materials gradient, adjusting the octahedral tilt synergistically to stabilize the lattice of the black-phase CsPbI₃ perovskite.¹³⁹ The presence of Br⁻ in crystal lattice is beneficial for reducing the octahedral tilting.¹⁴⁰

[H1] Challenges and Outlook

In the past few years, halide perovskites have advanced as a promising candidate for various highly efficient optoelectronic applications. Such rapid development could be attributed to empirical device and material development, and the pioneering exploration of their photophysical properties. Investigations have suggested that manipulating octahedral structures may be a critical factor in enhancing the performance and stability of optoelectronic perovskite-based devices.^{19,113,133} For an emerging topic, we expect that future studies could lead to efficiency limits within the conventional theoretical framework, stabilize perovskite-based materials and devices, and enable their high performance for commercialization (Fig. 8). These studies might focus on aspects related to: novel strategies and techniques for the precise manipulation of octahedral units; a greater understanding on the dynamics and local nature of octahedral tilting as well as the influence of lattice dynamics on phase transitions. Further explorations are necessary related to the charge carrier dynamics in materials and devices, and a better understanding of the correlation between octahedral tilting and stability.

[H2] Future strategies for manipulating octahedral units

As reviewed above, a series of methodologies (composition engineering, growth kinetics-associated modification, and interfacial treatments) have been developed to regulate the tilt angle of octahedra in halides perovskites. However, undesired trade-offs are still inevitable in these approaches. For example, while ion combination in the material can control the octahedral units, potential component segregation can also deteriorate the device performance over a long period of time, even if the additive is initially stable. Therefore, more advanced manipulation methods must be explored in halide perovskite films for any desirable optoelectronic applications.

According to the ionic and soft features of perovskite materials, some existing methodologies could be the starting point to fully control octahedral units. Firstly, steric effects can adjust the energetic barrier of phase transitions in perovskites. For instance, three molecules with anchoring group, bulk alkyl chain and different steric effects have been used to stabilize CsPbI₃ inorganic perovskites, suggesting the potential of employing steric molecules as a template to manipulate octahedral tiling.¹⁴¹ The physicochemical mechanism of how these molecules impact octahedra is still unclear, which should be investigated further. In addition, it is also a potential solution to use post-treatment strategy though it is mostly used for the passivation of surface defects in perovskites. For example, using methylammonium pyridine-2-carboxylic (MAPyA) for post-treatment of CsPbI₃ perovskites, Fu et al. successfully harnessed PyA- to control octahedral units and inhibit phase transitions.¹⁴² While clues are present in the scientific literature from various groups, systematic experiments are highly

desired to draw generalized conclusions for octahedral manipulation techniques.

[H2] Manipulating octahedral units for desirable carrier dynamics and better device performance

As mentioned previously, the configuration of octahedral units has considerable influences on the charge carrier dynamics, demonstrating the necessity to make efficient use of them in the novel devices. Novel devices with all kinds of features have been proposed and fabricated in order to chase and go beyond the SQ limit, but the structural trait of octahedral units opens up a distinct avenue for perovskite to transcend any theoretical limit.

We discussed that the control of octahedral units in perovskites enable a slowing down of carrier thermalization processes, inducing high carrier temperatures through the modulation of phonon modes among the crystal lattices. This highlighted new opportunities for controlling the charge carrier temperature without confinement systems, wherein spatial confinements usually require more efforts to transport carriers in the active media. It should be noted that various characterizations like neutron scattering, transient absorption spectroscopy, and temperature-dependent Raman spectroscopy have been used to investigate the carrier cooling process and depict the octahedral tilting in materials. Studies are needed to understand the configuration of carriers during the operation and the interaction between the lattice and carriers when a device outputs electricity under optical stimuli. Lastly, it is also crucial to design and construct devices with high carrier temperatures to increase the performance towards or even beyond the SQ limit.

[H2] Complete experimental understanding of the dynamic and local nature of octahedral tilting

The internal connection between octahedral tilting, lattice dynamics, and carrier dynamics in halide perovskites may enable the design of new devices. For instance, octahedral tilt-controlled carrier cooling processes are a promising way for hot-carrier PV devices with nearly doubled efficiency limits compared with conventional PV technologies. At the same time, non-radiative channels like Auger process can also be suppressed or made beneficial by modulating the intraband structure through altering the octahedral orientation. Such devices are far beyond the current techniques in optoelectronic areas. Currently, the vast majority of work concerned with these topics is computational, and this is an area not yet fully understood by the community. Typically, investigations on octahedral tilting requires experimental techniques — synchrotron or neutron sources and potentially energy-resolved measurements. Specific methodologies including both theoretical and experimental methodologies should be developed to reveal the effect of octahedral tilting on the lattice dynamics. Future work with time-resolved vibrational spectroscopy to investigate the structural dynamics of the perovskite lattice following photoexcitation may allow us to develop design rules for manipulating the structural dynamics that underpin the carrier and ion transport properties of the material. Although some research efforts are focused on the mechanistic studies, the more advanced techniques such as in-situ characterization could further deepen the fundamental understanding on the unique lattice dynamics and physical properties of halide perovskites. Furthermore, it is worth extending research from 3D perovskites to lower dimensional perovskites, to clarify the correlation between structural variation and optoelectronic processes in the materials.

[H2] Mechanism of octahedral units impacting the stability of halide perovskites

Phase transitions and decomposition are the two main degradation pathways for halide perovskite materials. For devices, the stability of the interfaces between perovskites and charge transport layers is also vital to understand and desired to be further improved. The control of octahedral units not only benefits the device performance, but also limits the degradation of the perovskite material and improves the long-term stability. On the one hand, both of the degradation pathways are thermodynamical processes involving the collapse of octahedra from the corner-sharing structure to the line-sharing or face-sharing configurations, leading to the demand of resilient metal-halide frameworks for better stability. The variation of tilting in octahedral units affects the carrier-phonon coupling, and therefore tunes the carrier dynamics.

Theoretical work by Eames predicted that tilting octahedra through mixed A-site cations may reduce lattice fluctuations in halide perovskites.⁴⁷ The reduced lattice fluctuations may decrease electron–phonon coupling within the material, reducing ion migration and contributing to an improved stability of halide perovskites.¹⁴³⁻¹⁴⁵ For further research in this area, in-situ techniques should be developed to correlate carrier-lattice interactions (such as polarons and carrier-phonon couplings) in the operating devices, so that the influence of electrical conditions can be included due to the ionic feature of halide perovskites. In addition, the reactivation or recovery of octahedral tilting in perovskites is important for maintaining the operational state of devices so as to prolong the lifetime of the perovskite materials.

[H1] Conclusion

In this work, we present the concept regarding octahedral units within halide perovskites and discuss their profound influence on the crystal structure, photophysical properties, and device performance. Our discussion also encompasses an overview of the methods used to characterize and manipulate octahedral tilting. Furthermore, we present a perspective on the future of research in this field, particularly for a deeper experimental understanding of the dynamic and local nature of octahedral tilting, which is essential for developing novel and efficient optoelectronic devices. Through understanding the nuances of octahedral tilting, researchers can strategically manipulate these octahedral units to tailor the properties of halide perovskites for developing specific applications beyond solar cells. We highlight the potential for exploring novel device designs beyond current applications that leverage octahedral tilting. For example, an understanding of the effect of octahedral tilting on charge carrier dynamics can potentially pave the way for novel, highly-efficient PV and other advanced optoelectronic technologies, and will lead to advances in materials science, optoelectronics, nanotechnology, and other fields.

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Author contributions

Yong W. conceived the idea. D. Yang, F. Gao, and S. D. Stranks supported this work. Yong W., Yu. W. and T. A. S. Doherty, wrote the original draft, all the authors reviewed and edited the draft.

Competing interests

The authors declare no competing interests.

Display items:

Tables:

Table 1 | A summary of the advantages and limitations of different characterisation techniques for probing octahedral tilting.

	Characterization	Advantages	Limitation
Diffraction techniques	X-ray diffraction	Quantitative information	Low sensitive
	Neutron diffraction	Quantitative Sensitive	Difficult to prepare sample
	Inelastic scattering	Local or macroscale, lattice dynamics	Difficult to prepare sample
	Electron diffraction	Sensitive	Complex multiple scattering. Dynamical interaction between scattered and incident electron waves.
Atomic resolution imaging	(Scanning) Transmission electron microscopy	Directly imaging	Challenge to resolve octahedral tilts of small magnitude, high electron dose
Other techniques	Nuclear magnetic resonance	Resolving octahedral tilting and phase transition	May not resolve subtle signs of symmetry breaking
	Optical spectroscopy	Convenient	Indirectly and cannot reflect the reality octahedral tilting

Figures:

Fig. 1 Crystal structures and octahedral tilting in halide perovskites. (a) Schematic illustration of the crystalline of hybrid perovskites of the formula ABX_3 , as well as the principal rotations of $CH_3NH_3^+$ (MA^+) and $CH(NH_2)_2^+$ (FA^+). Reproduced with permission.²⁰ Copyright. Annual Reviews publishing group. (b) Schematic diagram of 3D rotation of octahedra. (c) Effect of A-site cation on the octahedral tilting in halide perovskites. (d) The stable regime of geometrically stable cubic-like structure of three-dimensional (3D) halide perovskites, chemical limit (CL), octahedral limit (OL), stretch limit (SL), and tilt limits (TL).

Fig. 2 Examples of data extracted from perovskite materials that show the presence of octahedral tilting. (a) Maps of diffuse scattering in cubic $CsPbBr_3$ due to octahedral tilting. $K < 0$ shows Neutron data and $K > 0$ shows X-ray diffraction data. Reproduced with permission.⁵¹ Copyright 2021, Nature Publishing Group. (b) **(left to right)** TEM image of a $FAPbI_3$ perovskite film with 10mol% $CsCl$

addition that was indexed to an Im-3 crystal structure. Corresponding experimental Electron diffraction (ED) pattern extracted from the grain indicated in the transmission electron microscopy (TEM) image revealing an orientation near the [211] zone axis direction. A number of spots forbidden from appearing in the cubic Pm-3m phase are highlighted with red circles in the experimental ED pattern. Simulated kinematical ED pattern of the same orientation as the experimental pattern. Real space orientation of the perovskite grain of interest. Reproduced with permission.⁵⁹ Copyright 2021, Elsevier. (c) Atomic resolution high angle annular dark field STEM image of an LSMO/LFO heterostructure showing elliptical La columns in the LaFeO₃ layer (d) Quantification of the A-cation ellipticity as a function of position using 2D Gaussian fitting of the HAADF image (e) Annular bright field image of same region as c) (f) quantification of O shifts away from the position in a primitive perovskite cell due to octahedral tilting. Reproduced with permission.¹⁴⁶ Copyright 2019, American Physical Society.

Fig. 3 Dynamic B-cation off-center displacement and lattice anharmonicity in halide perovskites.

(a) Schematic diagram of B-site cation off-center in Pb-X octahedra. (b) Schematic illustration of various energy potentials from flat-bottom anharmonic to double well. Reproduced with permission.⁶⁸ Copyright 2021, Nature Publishing Group. Low-frequency Raman spectra of 3D perovskites CsPbBr₃ (c), MAPbBr₃ (d) and FAPbBr₃ (e) in the orthorhombic phase (blue), tetragonal phase (green) and cubic phase (red). Reproduced with permission. (c) and (d) are reprinted from ref.⁶⁹, Copyright 2017, American Physical Society. (e) is reprinted from ref.¹⁴⁷, Copyright 2020, American Chemical Society.

Fig. 4 Correlation between octahedral tilt and bandgaps. (a) Calculated band structures under biaxial

tensile, zero, and compressive strains from first-principle density functional theory (DFT) based approaches. Reproduced with permission.⁷⁵ Copyright 2019, Nature Publishing Group. (b) Schematic representation of bonding and antibonding orbitals of APbX₃ showing the formation of the valence band (VB) and conduction band (CB). Reproduced with permission.¹⁴⁸ Copyright 2015, Cambridge University Press. (c) The effect of octahedral rotation modes on the bandgap of CsPbI₃ perovskite, and (d) depict the bandgap shift due to the Pb site off-center along (100), (110), and (111) directions. Reproduced with permission.⁷⁸ Copyright 2021, Elsevier.

Fig. 5 Effect of octahedral variations on carrier dynamics. (a) Schematic illustration of different carrier cooling processes w/ and w/o octahedral tilts. (b) Strength of electron–phonon coupling of the MAPbI₃ phonon modes to the conduction band minimum (CBM) at the Γ point. Reproduced with permission.⁸¹ Copyright 2019, Wiley-VCH. (c) Schematic diagram of band structure splitting w/ and w/o octahedral tilts. (d) In the electron–electron–hole process, Auger coefficients versus scissors-shifted bandgap for the ideal and relaxed structures. Reproduced with permission.⁹² Copyright 2018, Wiley-VCH. (e) Illustrations for carrier scattering w/ and w/o octahedral tilts. (f) Spectral decomposition of the contribution of each phonon with energy $\hbar\omega$ to the electron scattering rate $1/\tau_{nk}$ in CsPbI₃. Reproduced with permission.⁹⁷ Copyright 2019, American Chemical Society.

Fig. 6 Tilting octahedra stabilizing halide perovskites. (a) and (b) Tilting [PbI₆]^{4−} octahedra stabilize FAPbI₃ perovskites. Reproduced with permission.¹⁹ Copyright 2021, American Association for the Advancement of Science (AAAS). (a) Progression of the structural transition from the corner-sharing tetragonal *P4/mbm* structure to the 2H hexagonal face-sharing structure by way of a mixed corner-

sharing and face-sharing intermediate structure. (b) The relative energy difference between the corner-sharing, mixed, and face sharing phases for the cubic-to-hexagonal and tetragonal-to-hexagonal transition. (c) Calculated stabilization energies as a function of the $(t+u)^n$ for halide perovskites with different octahedral tilting, tilted octahedra stabilize photoactive perovskites against degradation. Reproduced with permission.¹¹¹ Copyright 2018, American Chemical Society. (d) Introducing a steric molecule into γ -CsPbI₃ to suppress octahedral tilting. Reproduced with permission.¹¹⁵ Copyright 2017, Elsevier. (e) GIWAXs data for the β -CsPbI₃ with slight octahedral tilting. Reproduced with permission.¹¹³ Copyright 2019, AAAS.

Fig. 7 Manipulating octahedral units in the halide perovskites. (a) Schematic diagram of the halide perovskite structure without octahedral tilting. (b) Composition engineering regulates octahedral tilting in halide perovskites. (c) The incorporation of EA cation into 2D perovskite impacts the octahedral tilting. Reproduced with permission.¹²⁷ Copyright 2020, American Chemical Society. (d) growth kinetics-associated modification and (f) Interface temptation to induce the desirable octahedral tilting in halide perovskites.

Fig. 8 Research status and future outlook of octahedral unit in halide perovskites. A deeper experimental understanding of the dynamic and local nature of octahedral tilting is essential for developing novel and efficient optoelectronic devices. Through understanding the natures of octahedral tilting, these octahedral units can be strategically manipulated to tailor the properties of halide perovskites for developing specific applications beyond solar cells.

40-Word Summary

The octahedral unit is the fundamental and functional component of halide perovskites. The comprehensive and in-depth understanding of octahedral unit behavior will facilitate modulation of structure at the atomic level, benefiting perovskite (opto)electronic devices with enhanced efficiency, stability and functionality.

Fig 1

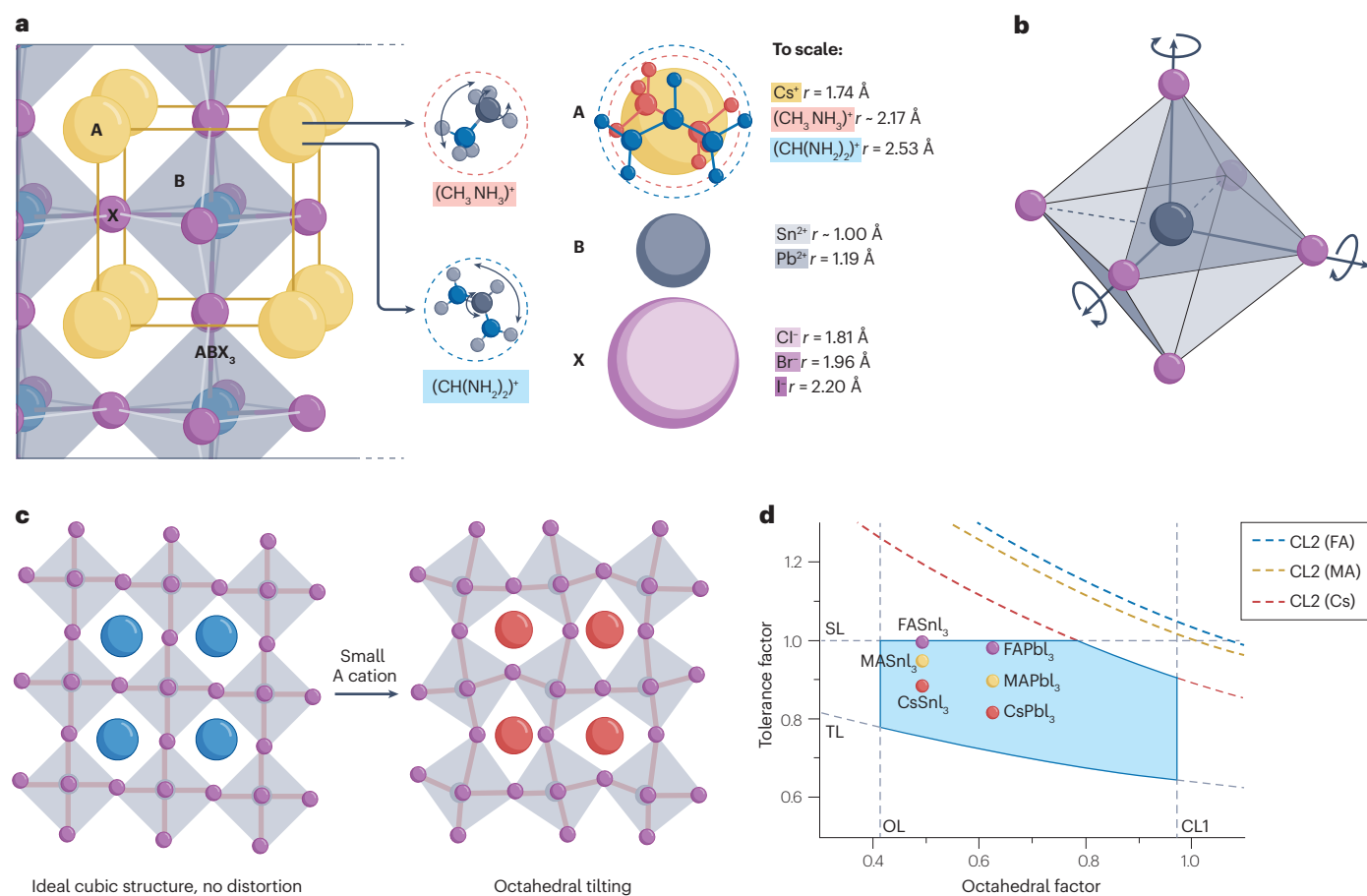


Fig 2

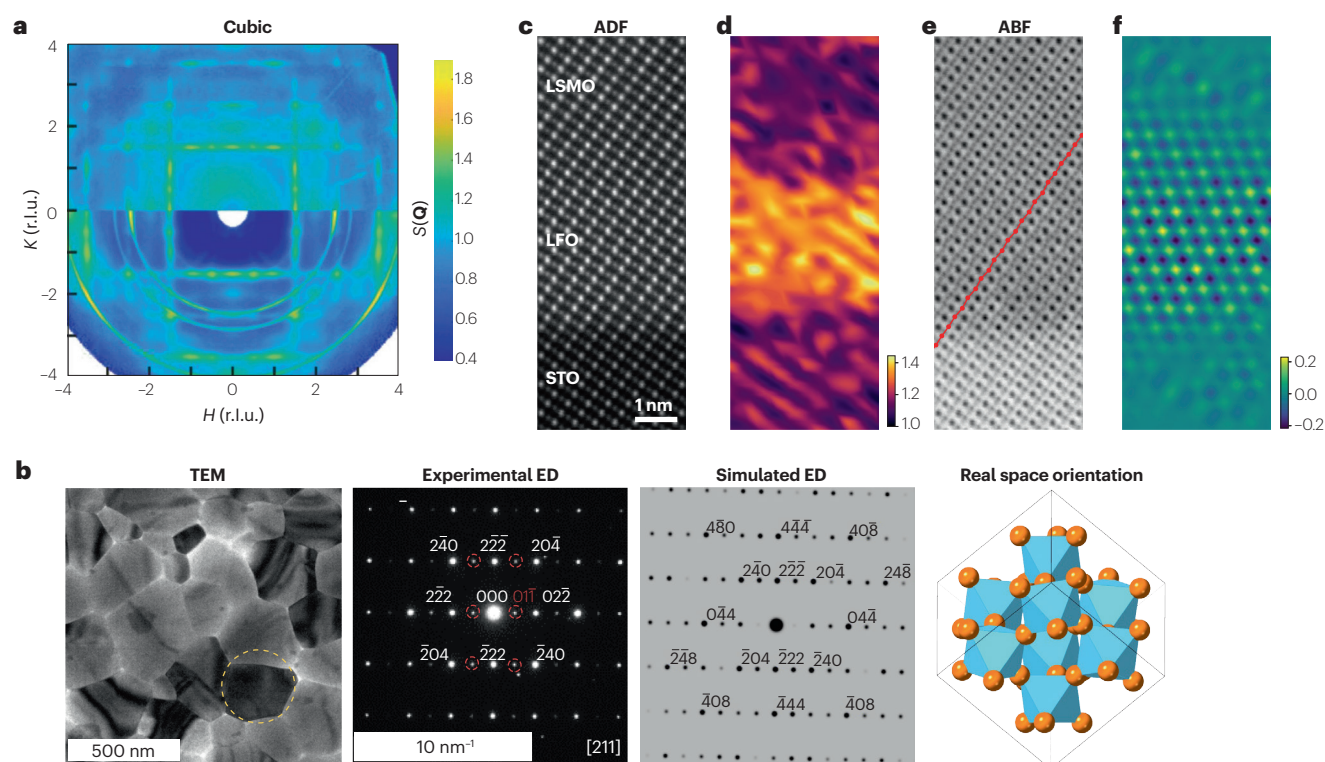


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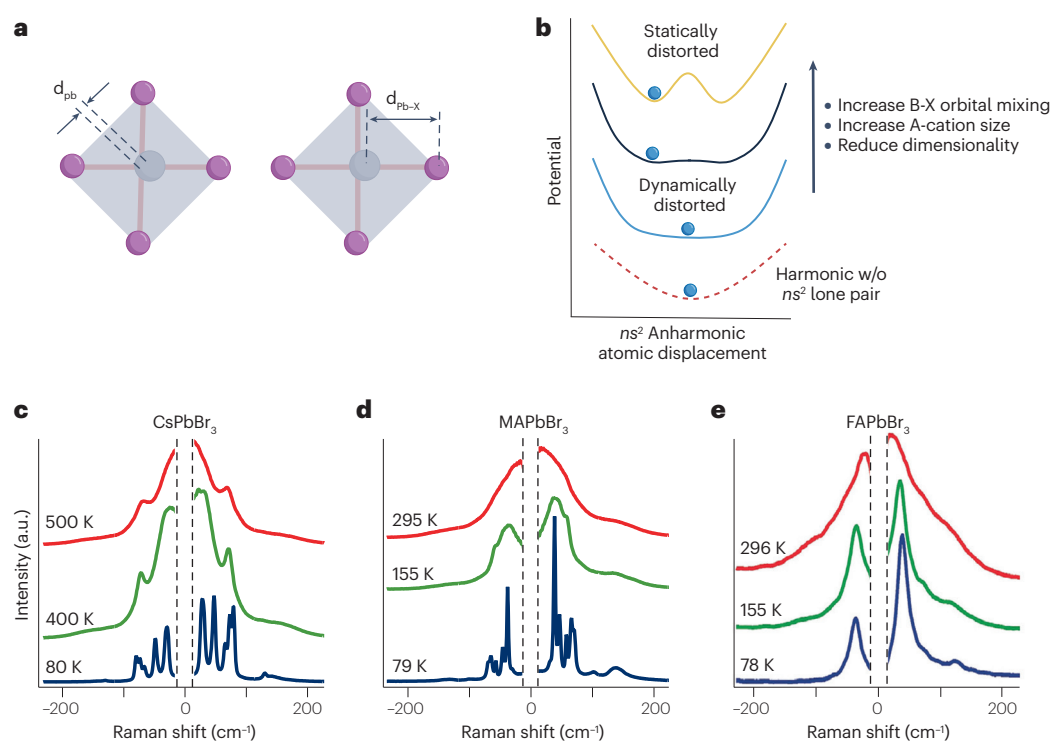


Fig. 4

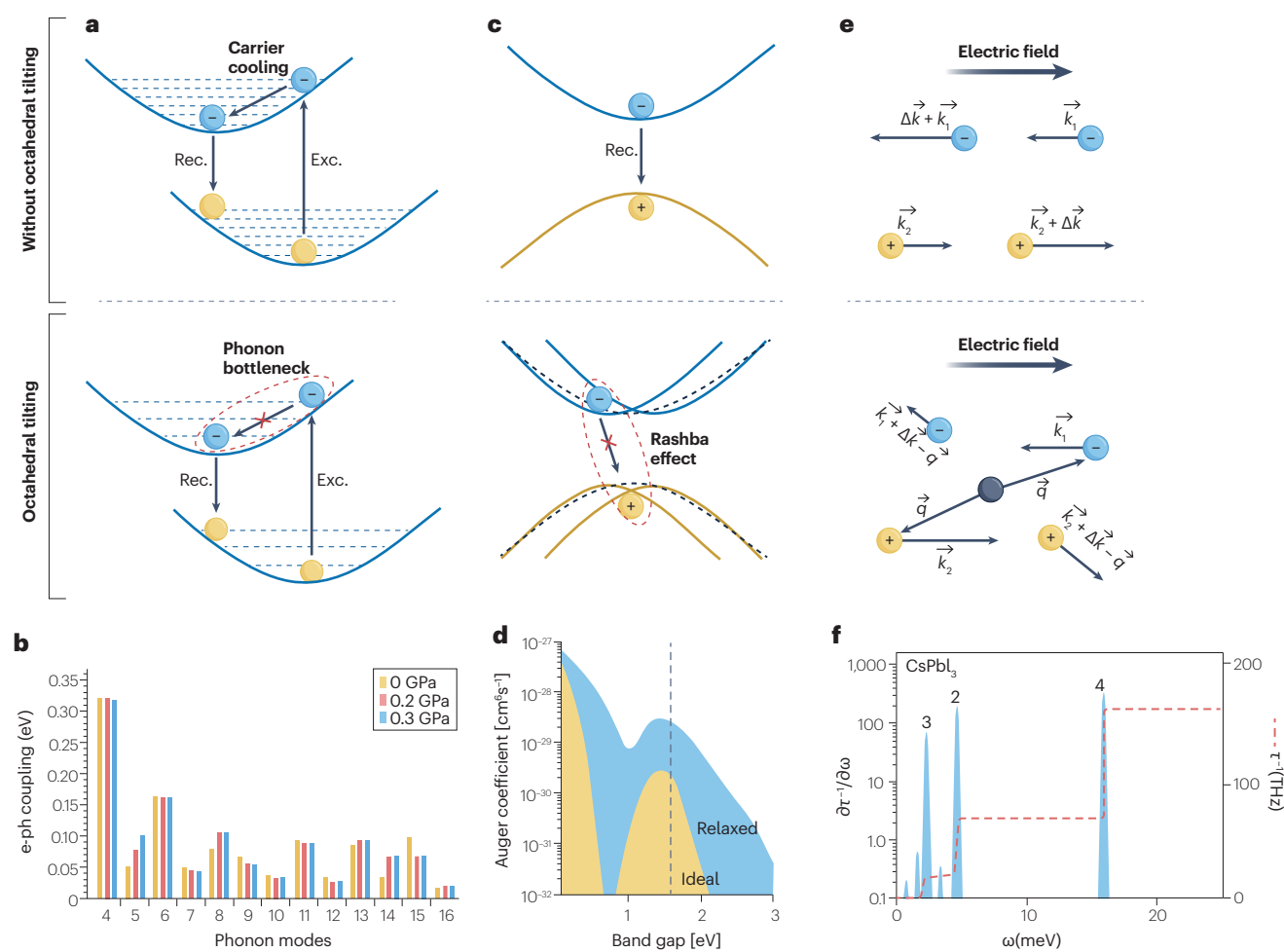


Fig 5

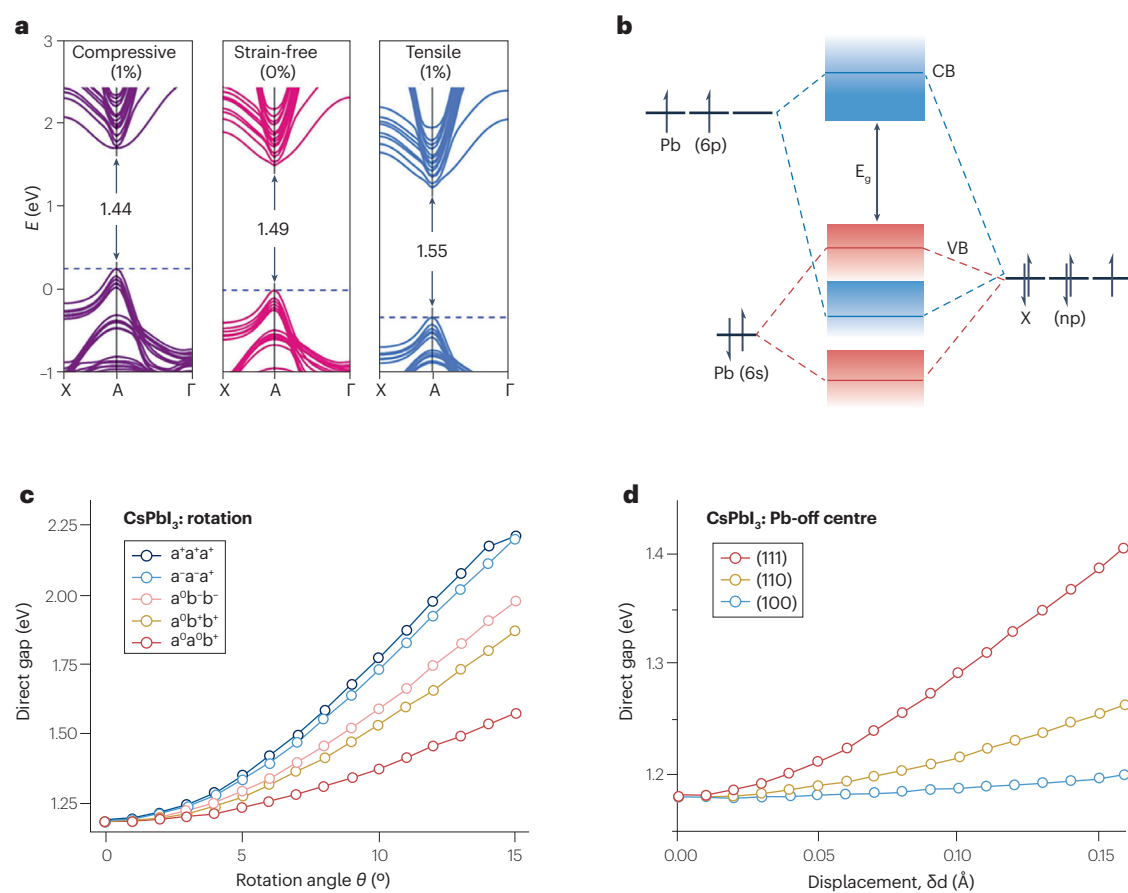


Fig 6

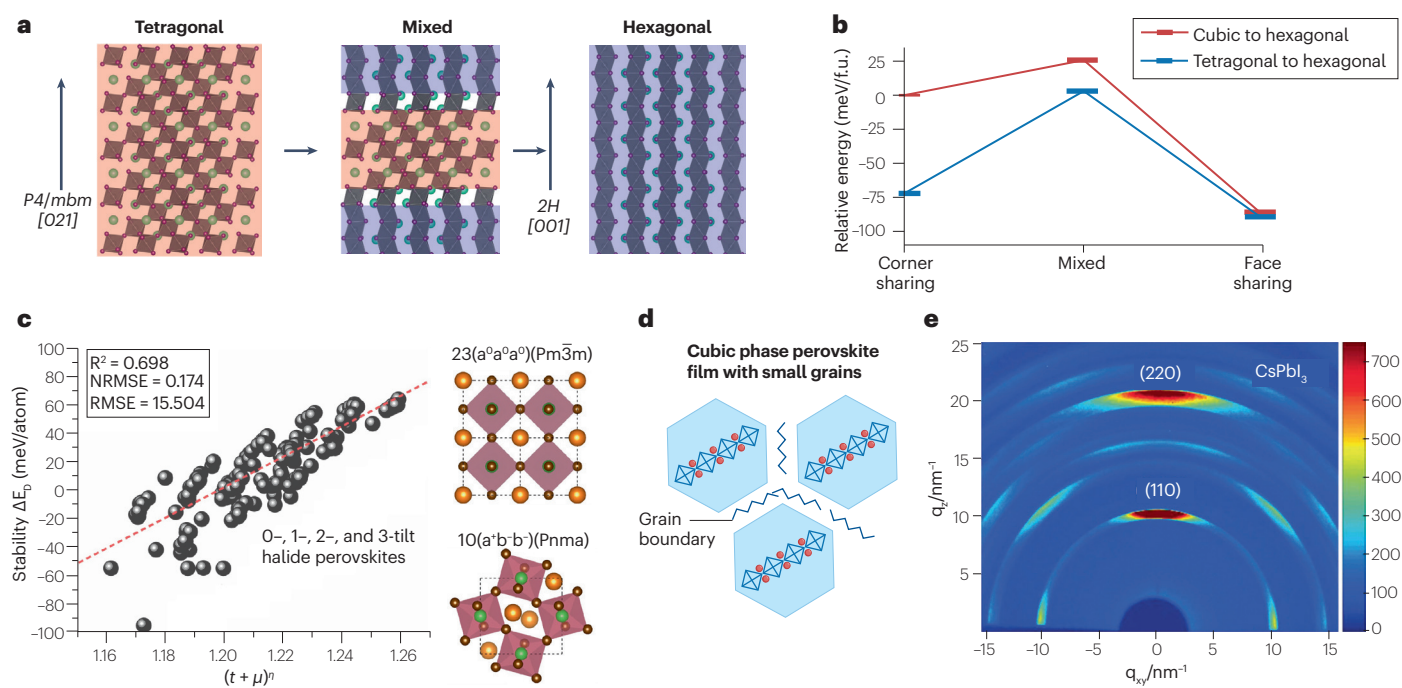


Fig 7

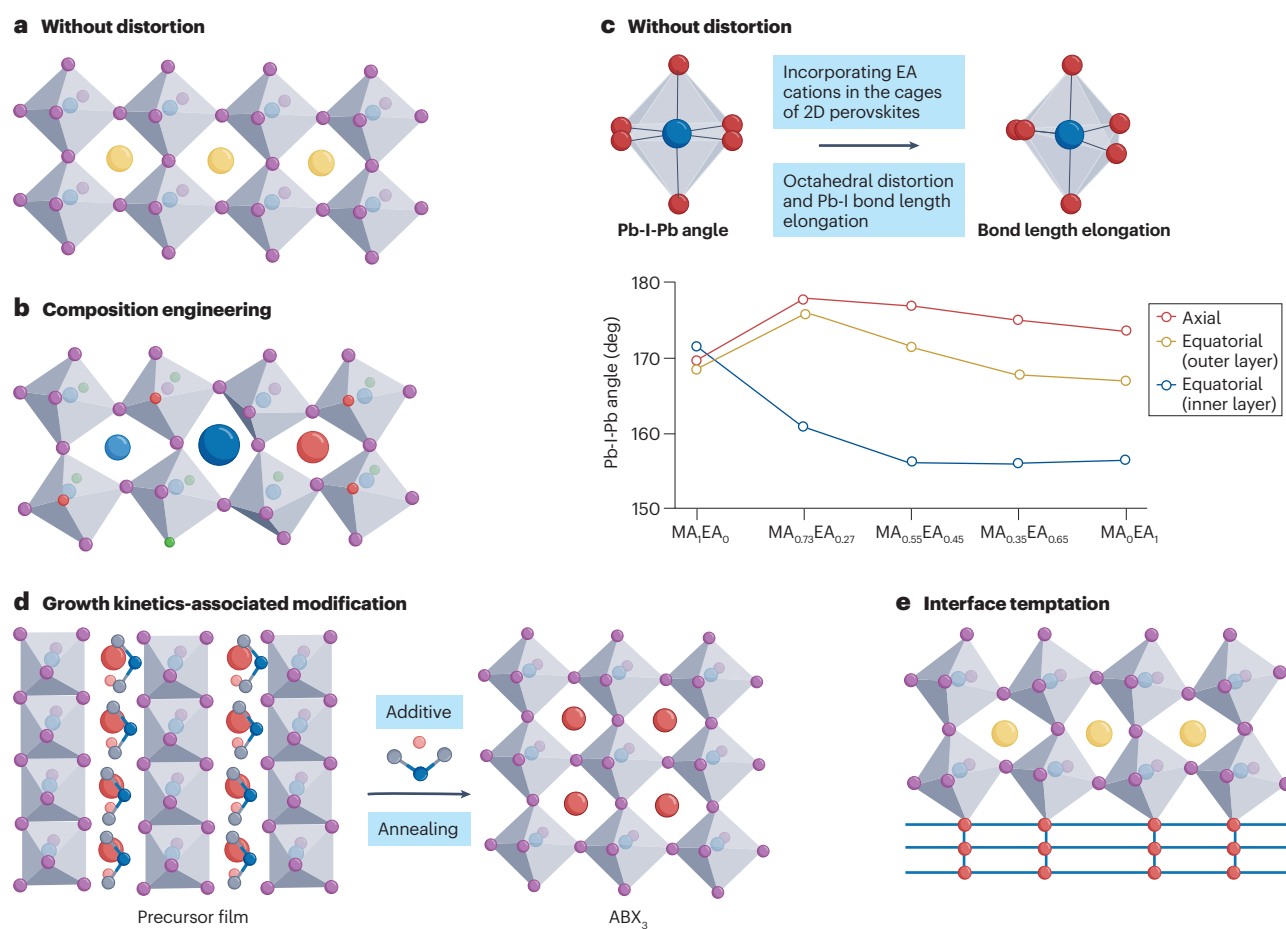
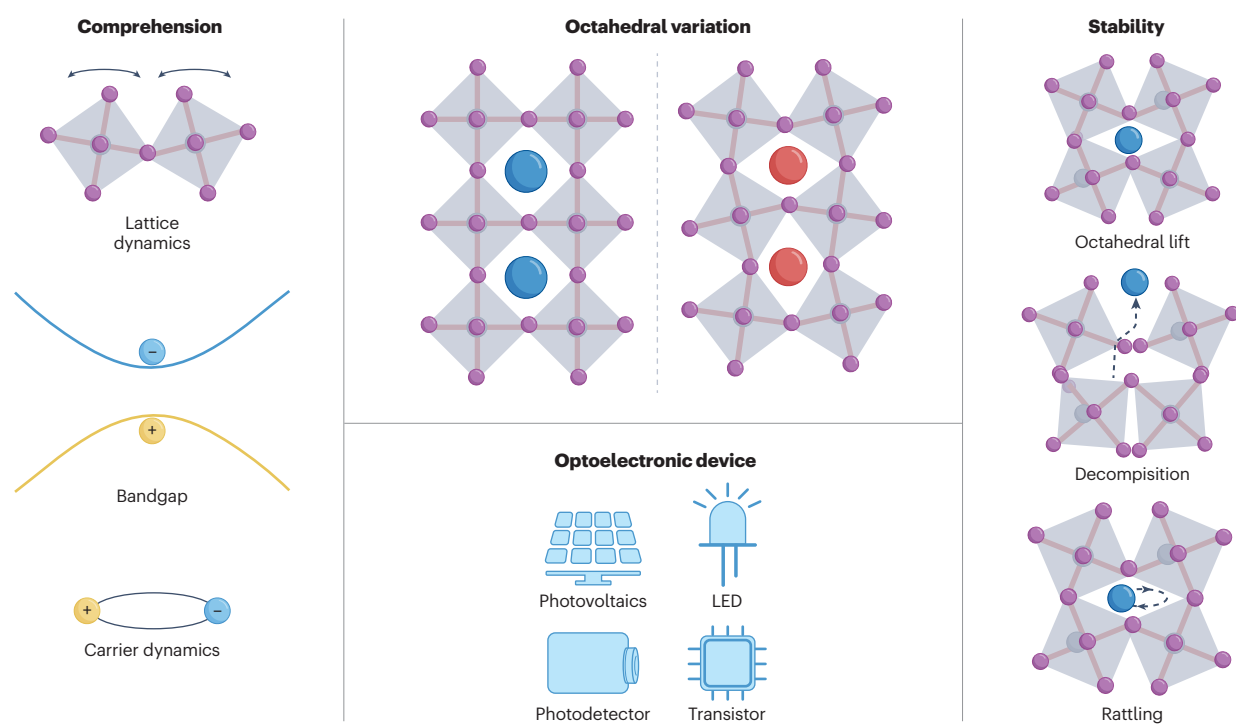


Fig 8



GA

